

Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term

Le Chen
Auburn University
le.chen@auburn.edu

Nicholas Eisenberg
Auburn University
nze0019@auburn.edu

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Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ for a Lipschitz continuous diffusion coefficient b which is not necessarily bounded. Using the theory of the stochastic integral laid out by John Walsh, we provide conditions which will guarantee the existence of an invariant measure for a broad range of initial conditions which includes L^2_ρ as well as the Dirac delta distribution δ_0 .

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1 Introduction

We consider the following *stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot), \end{cases} \quad (1.1) \quad \boxed{\text{E:SHE}}$$

where for some constants $L_b > 0$ and $L_0 \geq 0$, the function $b(x, u)$ satisfies the following Lipschitz and boundedness conditions:

$$|b(x, u) - b(x, v)| < L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R}, x \in \mathbb{R}^d. \quad (1.2) \quad \boxed{\text{E:lipcon}}$$

In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to the *parabolic Anderson model* [CM94]. In (1.1), $\dot{W}(t, s)$ is a centered Gaussian noise that is white in time and homogeneously correlated in space:

$$\mathbb{E} \left[\dot{W}(\psi) \dot{W}(\varphi) \right] = J(\psi, \varphi) := \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\varphi}(s, \cdot))(x), \quad (1.3) \quad \boxed{\text{E:Cor}}$$

where $\tilde{\varphi}(x) := \varphi(-x)$ and Γ is a tempered measure on $\mathbb{R}^d$ that is nonnegative and nonnegative definite, ϕ and φ are test functions, i.e., functions in $C_c^\infty(\mathbb{R}^{d+1})$. Γ is called the *correlation measure* and when it is absolutely continuous with respect to the Lebesgue measure, i.e., $\Gamma(dx) = f(x)dx$, we call f the *correlation function*. The Fourier transform of Γ (in the generalized

sense) is also a nonnegative and nonnegative definite tempered measure, which is called *spectral measure* and denoted by $\widehat{f}(d\xi)$. The initial condition μ is a deterministic, locally finite, regular, signed Borel measure with the corresponding *Hahn decomposition* being $\mu = \mu_+ - \mu_-$. Denote $|\mu| = \mu_+ + \mu_-$. The aim of the paper is to investigate the conditions required to guarantee the existence of invariant measures for the solution to (1.1), which is a crucial step towards the study of the ergodicity of the system.

We first note that the solution to (1.1) is interpreted as the *mild solution* as

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy) \quad (1.4) \quad \boxed{\text{E:gmsol}}$$

where $G(t, x) = (2\pi t)^{-d/2} \exp(-|x|^2/2t)$ is the heat kernel and the stochastic integral above is the *Walsh integral* (see [Wal86; Dal+09]). Throughout the paper, we use the following notation:

$$\begin{aligned} J_0(t, x) &:= \int_{\mathbb{R}^d} G(t, x - y) \mu(dy), & J_+(t, x) &:= \int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy), \\ \text{and } I(t, x) &:= \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy). \end{aligned} \quad (1.5) \quad \boxed{\text{E:J}}$$

The question of the existence of invariant measures in unbounded domains has been studied in few papers such as [AM03; MSY20; MSY16; TZ98]. Since the spatial domain is unbounded, one needs to introduce a weight function ρ and the corresponding weighted space $L_\rho^2(\mathbb{R}^d)$ as usual; see Section ?? below for more details. Following the standard procedure as laid out by Da Prato and Zabczyk [DGZ92], the main effort is to show that

$$\sup_{t>0} \mathbb{E} \left(\|u(t, \cdot)\|_{L_\rho^2(\mathbb{R}^d)}^2 \right) < \infty. \quad (1.6) \quad \boxed{\text{E:BddMnt}}$$

For convenience, we will write $\|\cdot\|_\rho$ instead of $\|\cdot\|_{L_\rho^2(\mathbb{R}^d)}$.

Recall that under the following Assumption 1.1, one can find a unique mild solution to (1.1) with sharp moment estimates; see [CK19; CH19; CD15].

Assumption 1.1. Random initial measure. Need to clarify this.

- (i) The initial measure μ satisfies the following integrability condition:

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) \mu_2^*(dx) < \infty \quad \text{for all } a > 0, \quad (1.7) \quad \boxed{\text{A:icon}}$$

where μ_*^2 is defined below in (1.12).

- (ii) The spectral measure $\widehat{f}$ satisfies *Dalang's condition*, namely,

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{\beta + |\xi|^2} < \infty \quad \text{for some and therefore any } \beta > 0. \quad (1.8) \quad \boxed{\text{A:dalcon}}$$

I think (1.7) is good enough

Moreover, this solution is usually *intermittent*, namely, the probability moments to (1.1) have certain exponential growth in t . In order to have moments bounded in time as required in (1.6), one has to first identify the exact conditions, under which the moments to (1.1) do not possess exponential growth. This question has been answered in [CK19], where necessary and sufficient conditions are given. In [AM03; MSY20; MSY16], a drift term $f(x, u)$ is included in (1.1) in a crucial way to help to cancel the otherwise exponential growing moments. The absence

of such a drift term in this paper makes the problem more challenging. According to part (3) of Theorem 1.3 of [CK19], in order to have (1.6), one has to have L_b in (1.2) — the Lyapunov exponent of b — small enough and

$$\Upsilon(0) < \infty, \quad (1.9) \quad \boxed{\text{E:U0finite}}$$

where $\Upsilon(\cdot)$ is defined in (1.8). Note that condition (1.9) is equivalent to

$$d \geq 3 \quad \text{and} \quad \int_{\mathbb{R}^d} \frac{1}{|z|^{d-2}} \Gamma(dz) < \infty, \quad (1.10) \quad \boxed{\text{E:U0Eqv}}$$

where Γ is the correlation measure (see (1.3)). Condition (1.9) (or equivalently (1.10)) gives the admissible correlation functions, which is weaker than that in [TZ98]; see Remark 1.2 below.

(3): Now that we have condition (1.11) written correctly, I believe that (1.9) implies (1.11). We consider

$$\Upsilon(0) = \int_{\mathbb{R}^d} \frac{\hat{f}(z)}{|x|^2} dz < \infty.$$

It seems that the worst function that satisfies this is

$$\hat{f}(z) = \begin{cases} \frac{1}{|z|^{d-2-\delta}} & |z| \leq 1 \\ \frac{1}{|z|^{d-2+\eta}} & |z| \geq 10 \\ \text{continuous} & 1 \leq |z| \leq 10, \end{cases}$$

where $\eta > 0$ and $\delta > 0$ are arbitrary. We want to show that such a $\hat{f}$ also satisfies (1.11). Note that we need the following:

$$p(d-2-\delta) < d \quad \text{and} \quad p(d-2+\eta) > d.$$

This implies that

$$\frac{d}{d-2+\eta} < p < \frac{d}{d-2-\delta},$$

and note that for any $\eta > 0$, that

$$\frac{d}{d-2+\eta} < \frac{d}{d-2},$$

which implies

$$\left(\frac{d}{d-2+\eta}, \frac{d}{d-2-\delta} \right) \cap \left[1, \frac{d}{d-2} \right) \neq \emptyset.$$

Therefore there will always be a $p \in [1, d/(d-2))$ such that $\hat{f}$ satisfies (1.11). Hence (1.11) is more general than (1.9).

Did you try the reverse order? (1.11) implies (1.9)?

On the discussion file on the master branch I have an example which works for (1.11) but not (1.9).

The issue of your arguments is that we may have logarithm corrections for $\hat{f}$, such as

$$\hat{f}(z) = \frac{1}{|z|^{d-2} \log(|z|) (1 + |z|^\delta)}$$

where $\delta > 0$. Certainly we need to make this nonnegative definite-definite, which should be doable. Indeed in my paper "Dense blowup...", we have constructed such correlation functions. You can play with this idea to include more logarithms such as

$$\hat{f}(z) = \frac{1}{|z|^{d-2} \log(|z|) \log(\log(|z|)) (1 + |z|^\delta)}.$$

You can also make the logarithm correction at infinity not zero. So my impression is that the integrability condition of type (1.11) might not be the best way to characterize the situation.

R:Cor

Remark 1.2. The assumption on the correlation function in [TZ98] is given through the spectral density $\widehat{f}$ as follows:

$$\widehat{f} \in L^p(\mathbb{R}^d) \quad \text{for some } p \in [1, d/(d-2)); \quad (1.11)$$

E:CorTZ

see Hypothesis 2.1 (*ibid.*) and recall that $d \geq 3$. One can easily check if $\widehat{f}$ satisfies (1.9), then one can find a p in the range given in (1.11) such that condition (1.11) is satisfied. In some sense, our condition (1.9) seems to be more precise and convenient to check. **Need to be extended...**

Notation 1.3. We need to introduce some notation. Denote

$$J_q^*(r, z) = (G(r, \cdot) * \mu_q^*)(x) \quad \text{and} \quad \mu_q^* := 1 + \|\mu\|_q. \quad (1.12)$$

E:qJ*

We make note that for deterministic initial measures, that the above just simplifies to

$$J^*(r, z) = (G(r, \cdot) * \mu^*)(x) \quad \text{and} \quad \mu^* := 1 + |\mu| \quad (1.13)$$

E:J*

T:Lrho

Theorem 1.4. Assume that

(i) $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is an nonnegative $L^1(\mathbb{R}^d)$ function;

(ii) the initial measure μ , either random or deterministic, satisfies that

$$\mathcal{G}_{\rho,2}(t) = \mathcal{G}_{\rho,\mu_2^*}(t) := \int_{\mathbb{R}^d} (J_2^*(t, x))^2 \rho(x) dx < \infty \quad \text{for all } t > 0; \quad (1.14)$$

E:G

(iii) the correlation measure Γ satisfies (1.9) (or equivalently (1.10));

(iv) the Lipschitz constant L_b of b (see (1.2)) satisfies

$$L_b < \quad (1.15)$$

E:LipLb

Then for all $t > 0$, $u(t, \cdot) \in L_\rho^2(\mathbb{R}^d)$ a.s. Moreover, for some constant $C > 0$ which does not depend on t , it holds that

$$\mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) \leq C \mathcal{G}_{\rho,2}(t) < \infty. \quad (1.16)$$

E:uInLrho

Proof. Under conditions (iii) and (iv), we can apply the moment bound (2.7) to see that for all $t > 0$,

$$\begin{aligned} \mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) &= \mathbb{E} \left(\int_{\mathbb{R}^d} |u(t, x)|^2 \rho(x) dx \right) \\ &\leq C \int_{\mathbb{R}^d} \left(1 + (\|\mu\|_2 * G(t, \cdot))(x) \right)^2 \rho(x) dx \\ &= C \int_{\mathbb{R}^d} \left((G(t, \cdot) * (1 + \|\mu\|_2))(x) \right)^2 \rho(x) dx \\ &= C \int_{\mathbb{R}^d} J_2^*(t, x)^2 \rho(x) dx \\ &= C \mathcal{G}_{\rho,2}(t) < \infty \end{aligned}$$

where we have applied Fubini's Theorem and the constant C does not dependent on t . This proves Theorem 1.4. $\square$

Here is the main theorem of the paper:

T:Main

Theorem 1.5. Assume that all assumptions of Theorem 1.4 hold. Moreover, suppose that

(i) ρ be an admissible weight and there exists another admissible weight $\widehat{\rho}$ such that

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty; \quad (1.17) \quad \text{E:rho rho}$$

(ii) the weight function $\widehat{\rho}$ and the initial condition satisfy the following condition:

$$\limsup_{t>0} \mathcal{G}_{\widehat{\rho},q}(t) < \infty; \quad (1.18) \quad \text{E:InitData}$$

(iii) for some $\alpha \in (0, 1/2)$, the spectral measure $\widehat{f}$ satisfies that

$$\mathcal{H}_\alpha(t) := \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) < \infty \quad \text{for all } t > 0; \quad (1.19) \quad \text{E:H}$$

(iv) for some $q > 2$ and $\alpha \in (q^{-1}, 2^{-1})$,

$$\int_0^t [\mathcal{G}_{\widehat{\rho},q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty, \quad \text{for all } t > 0. \quad (1.20) \quad \text{E:GH-int}$$

Then we have that

(a) for any $t_0 > 0$, the sequence of laws of $u(t, \cdot)\}_{t \geq t_0}$ is tight, i.e., for any $\epsilon \in (0, 1)$, there exists a compact set $\mathcal{K} \subset L_\rho^2(\mathbb{R}^d)$ such that

$$\mathbb{P}(u(t, \cdot) \in \mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq t_0; \quad (1.21) \quad \text{E:cpt}$$

(b) there exists an invariant measure for the laws of $\{u(t, \cdot)\}_{t>0}$.

Remark 1.6. We do not need to require that $u(t, \cdot) \in L_\rho^2(\mathbb{R}^d)$ in (1.21). But if (1.18) is satisfied with $\widehat{\rho}$ is replaced by ρ , which is the case, for example, when $\rho(x) \leq C \widehat{\rho}(x)$ for all $x \in \mathbb{R}^d$, then we do have $u(t, \cdot) \in L_\rho^2(\mathbb{R}^d)$ and $\sup_{t>0} \mathbb{E}(\|u(t, \cdot)\|_\rho^2) < \infty$ by part (a) of Theorem 1.5.

Need to make a comment that when $u(0, \cdot) = \delta_0(\cdot)$ then $u(t, \cdot) \in L_\rho^2$ for any weight ρ . This is important for (3.14).

E:Gamma

Example 1.7 (Correlation measures). Here are a few examples of correlation measures Γ that satisfy condition (1.20). The Bessel potential (see, e.g., Section 1.2.2 of [Gra14])... ... TBD ...

E:Init

Example 1.8 (Initial conditions). Here are few examples:

1. If the initial condition μ is the Lebesgue measure, than the left-hand side of (1.18) is equal to $\|\widehat{\rho}\|_{L^1(\mathbb{R}^d)}$, which is finite and does not depend on t .
2. If $\mu = \delta_0$,

Proof of Theorem 1.5 (b). Fix an arbitrary $T_0 > 0$ and denote

$$U(T) := \frac{1}{T} \int_{T_0}^{T+T_0} \mathcal{L}(u(t, \cdot)) dt, \quad T > 0.$$

We claim that the family of laws $U(T, \cdot)$ for $T > 0$ is tight in $L^2_\rho(\mathbb{R}^d)$. Indeed, for any $\epsilon \in (0, 1)$, by part (a), there exists a compact set $\mathcal{K} \in L^2_\rho(\mathbb{R}^d)$ such that (1.21) holds, which implies that

$$U(T)(\mathcal{K}^c) = \frac{1}{T} \int_{T_0}^{T+T_0} \mathcal{L}(u_{u_0}(t, \cdot))(\mathcal{K}^c) dt \leq \frac{1}{T} \int_{T_0}^{T+T_0} \epsilon dt = \epsilon, \quad \text{for all } T > 0.$$

Let $\{T_n\}_{n \in \mathbb{N}}$ be any deterministic sequence such that $T_n \uparrow \infty$. Since $\{U(T_n)\}_{n \geq 1}$ is a tight sequence of measures, there exists a subsequence $\{U(T_{n_m})\}_{m \geq 1}$ that converges weakly to a measure, say, η , on $L^2_\rho(\mathbb{R}^d)$. Then one can apply for the *Krylov-Bogoliubov theorem* (see, e.g., [DZ96, Theorem 3.1.1]) to conclude that the measure η is the invariant measure for $\mathcal{L}(u(t, \cdot))$, $t \geq T_0$. Finally, we can choose T_0 to be arbitrarily close to zero. This completes the proof of part (c) of Theorem 1.5. $\square$

2 Preliminaries

Let ρ be some nonnegative function on $\mathbb{R}^d$. We denote by $L^2_\rho(\mathbb{R}^d)$ the weighted Hilbert space with the inner produce $\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x)dx$ and set $|f|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2\rho(x)dx$. Following Tessitore and Zabczyk [TZ98], we make the following definition:

D:Rho

Definition 2.1. $\rho : \mathbb{R}^d \mapsto \mathbb{R}$ is called an *admissible weight* if ρ is a strictly positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d, \quad (2.1)$$

E:aw

or equivalently,

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq C_\rho(T). \quad (2.2)$$

E:awbd1

Several examples of admissible weights are (see Example in Section 2 of [TZ98])

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \quad (2.3)$$

E:admRho

R:admRho

Remark 2.2. The smaller the weight function $\rho(\cdot)$ (not necessarily to be admissible here) is, the larger the space $L^2_\rho(\mathbb{R}^d)$ is. For example, one may choose ρ to be either a nonnegative function with compact support or the heat kernel itself $G(1, \cdot)$. In both cases, ρ is smaller than those in (2.3) (up to a constant). However, one can easily check that the admissible condition (2.1) excludes these two cases. It is much less obvious that the following weight function is not admissible:

$$\rho(x) = \exp(-a|x|^b), \quad x \in \mathbb{R}^d, \quad \text{with } a > 0 \text{ and } b \in (1, 2) \text{ fixed.}$$

P:weight

Proposition 2.3 (Proposition 2.1 of [TZ98]). *For any admissible weight ρ , the operators on $L^2_\rho(\mathbb{R}^d)$ defined by $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 -semigroup on L^2_ρ . Moreover, if $\widehat{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty,$$

then for any $t < 0$, the operators defined above are compact from $L^2_\rho(\mathbb{R}^d)$ to $L^2_{\widehat{\rho}}(\mathbb{R}^d)$.

We need Lemma 2.4 to how that $\varphi \in L^2_\rho$ will satisfy (1.7).

L:RhoDom

Lemma 2.4. *For any admissible weight function ρ and $a > 0$, there exists a constant $K_{\rho,a} > 0$ such that*

$$\exp(-a|x|^2) \leq K_{\rho,a} \rho(x), \quad \text{for all } x \in \mathbb{R}^d. \quad (2.4) \quad \text{E:RhoDom}$$

Proof. Let ρ be an arbitrary admissible weight function and $a > 0$. By (2.1),

$$\begin{aligned} \frac{\exp(-a|x|^2)}{\rho(x)} &= \frac{\exp(-a|x|^2)}{\rho(x)} \frac{(G(1, \cdot) * \rho(\cdot))(x)}{(G(1, \cdot) * \rho(\cdot))(x)} \\ &\leq \left(\sup_{x \in \mathbb{R}^d, t \in [0,1]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \right) \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)} \\ &\leq C_\rho(1) \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)}. \end{aligned}$$

Noticing that for some constant $\Theta > 0$, $\rho(x) \geq \Theta \prod_{\ell=1}^d \mathbb{I}_{[-1,1]}(x_\ell)$, a simple exercise shows that

$$C_{\rho,a} := \sup_{x \in \mathbb{R}^d} \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)} \leq \frac{1}{\Theta} \left(\sup_{x \in \mathbb{R}} \frac{\exp(-ax^2)}{(G(1, \cdot) * \mathbb{I}_{[-1,1]}(\cdot))(x)} \right)^d < \infty.$$

This proves the lemma with $K_{\rho,a} = C_\rho(1)C_{\rho,a}$. $\square$

Theorem 2.5 (Theorem 1.2 [CH19]). *Under Assumption 1.1, the SPDE (1.1) has a unique random field solution starting from μ and the solution is $L^2(\Omega)$ continuous.*

T:ExUn

Theorem 2.6 (Theorem 1.7 [CH19]). *Under Assumption 1.1, for any $t > 0$ and $x \in \mathbb{R}^d$, the solution to (1.1), $u(t, x)$, given by (1.4) is in $L^p(\Omega)$ and for $p \geq 2$*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(\|\mu\|_p * G(t, \cdot))(x)] H(t; \gamma_p)^{1/2}, \quad (2.5) \quad \text{E:mb}$$

where $\bar{\varsigma} = L_0/L_b$, $\gamma_p = 32pL_b^2$ and the function $H(t; \gamma_p)$ is nondecreasing in t (see [CH19] for the concrete definition of H). Moreover, if we assume that

$$\Upsilon(0) < \infty \quad (2.6) \quad \text{A:dalcon0}$$

then this function $H(t; \gamma_p)$ is bounded in t . Hence, by choosing $C = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$, we have that

$$\|u(t, x)\|_p \leq C(L_0 + J_+(t, x)), \quad (2.7) \quad \text{E:mb0}$$

where L_0 is from (1.2).

The following factorization property of the heat kernel will be used in this paper:

$$G(s - r, y - z)G(r, z - \sigma) = G(s, y - \sigma)G\left(\frac{r(s - r)}{s}, z - \sigma - \frac{r}{s}(y - \sigma)\right). \quad (2.8) \quad \text{E:gsim}$$

3 Proof of part (a) of Theorem 1.5

For $\alpha \in (0, 1)$, $t > 0$ and $x \in \mathbb{R}^d$, define formally

$$\begin{aligned} (F_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t - s)^{\alpha-1} G(t - s, x - y) f(s, y) \, ds dy \quad \text{and} \\ (Y_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t - s)^{-\alpha} G(t - s, x - y) f(s, y) W(ds, dy). \end{aligned} \quad (3.1) \quad \text{E:Ops}$$

For F_α , the Step 1 of the proof of [TZ98, Theorem 3.1] showed the following proposition:

P:F-Comp

Proposition 3.1. *Let ρ and $\hat{\rho}$ be given in condition (i) of Theorem 1.5 (see (1.17)). For any $q > 2$, $t_0 > 0$ and $\alpha \in (q^{-1}, 2^{-1})$, the operator F_α as an operator from $L^q((0, t_0); L_\rho^2)$ to $L_\rho^2(\mathbb{R}^d)$ is compact.*

As for Y_α , we have the following result:

L:yineq

Lemma 3.2. *For any $\alpha \in (0, 1/2)$, denote (with slightly abuse of notation)*

$$Y(s, y) := [Y_\alpha b(\circ, u(\cdot, \circ))](s, y) = \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, u(r, z)) W(dr, dz). \quad (3.2)$$

E:Y

Under the assumptions of Theorem 1.4 and (iii) of Theorem 1.5, then for all $q \geq 2$, $s > 0$ and $y \in \mathbb{R}^d$,

$$\|Y(s, y)\|_q^2 \leq C_q [J_q^*(s, y)]^2 \mathcal{H}_\alpha(s). \quad (3.3)$$

E:NormY

Moreover, if

(iv') *for some $q \geq 2$ and $\alpha \in (0, 1/2)$, the weight function ρ (not necessarily admissible)*

$$\int_0^t [\mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty, \quad \text{for all } t > 0, \quad (3.4)$$

E:GH-int'

then there exists a constant $\Theta = \Theta(q, L_b, L_0, \alpha)$ such that

$$\mathbb{E} \left(\int_0^t \|Y(s, \cdot)\|_\rho^q ds \right) \leq \Theta \int_0^t [\mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty. \quad (3.5)$$

E:yineq

Proof. In the proof, we use C to denote a generic constant that may change its value at each appearance. First, an application of Minkowski's inequality shows that

$$\mathbb{E} \left(\|Y(s, \cdot)\|_\rho^q \right) = \left\| \int_{\mathbb{R}^d} Y(s, y)^2 \rho(y) dy \right\|_{q/2}^{q/2} \leq \left(\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy \right)^{q/2}. \quad (3.6)$$

E:EYqr

Now we look to find an appropriate upper bound for $\|Y(s, y)\|_q^2$. By the Burkholder-Davis-Gundy inequality, as well as Minkowski's integral inequality to see that

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq C_q \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(z_1, u(r, z_1))\|_q f(z_1 - z_2) \\ &\quad \times G(s-r, y-z_2) \|b(z_2, u(r, z_2))\|_q. \end{aligned}$$

Note that for the Lipschitz condition in (1.2), we have that

$$|b(x, u)| \leq |b(x, u) - b(x, 0)| + |b(x, 0)| \leq L_b |u| + L_0 \leq C(1 + |u|), \quad C := L_b \vee L_0.$$

We apply this and the moment bound (2.7) to $\|b(z_i, u(r, z_i))\|_q$ above to see that

$$\|b(z_i, u(r, z_i))\|_q \leq C \left(1 + \|u(r, z_i)\|_q \right) \leq C \left(1 + (G(r, \cdot) * \|\mu\|_q)(z_i) \right) = C J_q^*(r, z_i), \quad i = 1, 2, \quad (3.7)$$

E:NormB

where J_q^* is defined in (1.12). Therefore,

$$\|Y(s, y)\|_q^2 \leq C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) J_q^*(r, z_i) \right)$$

$$\begin{aligned}
&= C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \prod_{i=1}^2 \left(G(s-r, y - z_i) G(r, z_i - \sigma_i) \right) \\
&= C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) G(s, y - \sigma_1) G(s, y - \sigma_2) \\
&\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i - \sigma_i - \frac{r}{s}(y - \sigma_i)\right) \\
&\leq C(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) G(s, y - \sigma_1) G(s, y - \sigma_2) \\
&\quad \times \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right),
\end{aligned}$$

where we have applied (2.8) and Plancherel's theorem. Therefore,

$$\|Y(s, y)\|_q^2 \leq C(2\pi)^{-2d} [J_q^*(s, y)]^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right). \quad (3.8) \quad \boxed{\text{E_ : NormY}}$$

By symmetry of $r(s-r)/s$ term and the fact that $r(s-r)/s \geq r/2$ for all $r \in [0, s/2]$, we see that the above double integral is bounded by

$$\begin{aligned}
&\leq 2 \int_0^{s/2} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2}|\xi|^2\right) \\
&= 2^{2(1-\alpha)} \int_0^{s/4} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) \\
&\leq 2^{2(1-\alpha)} \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) \\
&= 2^{2(1-\alpha)} \mathcal{H}_\alpha(s).
\end{aligned}$$

Plugging the above bound back to (3.8) proves (3.3).

It remains to prove (3.5). By the definition of $\mathcal{G}_\rho(t)$ in (1.14) and by (3.3), we see that

$$\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy \leq C \mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s).$$

Plugging the above expression to the far right-hand of (3.6) proves (3.5). Finally, the finiteness of the upper bound in (3.5) is guaranteed by condition (3.4). This completes the proof of Lemma 3.2. $\square$

L:Factorize

Lemma 3.3 (Factorization lemma). *Let $u(t, x)$ be the solution to (1.1) and Y be given in (3.2). Under the conditions of Theorem 1.4 and (iii) of Theorem 1.5, for all $\alpha \in (0, 1/2)$, the following factorization holds*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, u(r, z)) W(dr, dz).$$

As a consequence,

$$u(t, x) = J_0(t, x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y](t, x), \quad \text{for all } t > 0 \text{ and } x \in \mathbb{R}^d. \quad (3.9) \quad \boxed{\text{E:1fac}}$$

Proof. The lemma is straightforward provided that one can switch the orders of stochastic and ordinary integrals:

$$\begin{aligned}
& \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds \\
&= \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \\
&\quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, u(r, z)) W(dr, dz) \\
&= \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(z, u(r, z)) W(dr, dz) \quad (3.10) \quad \boxed{\text{E:Fubini}_1}
\end{aligned}$$

$$\begin{aligned}
&= \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(u(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \quad (3.11) \quad \boxed{\text{E:Fubini}_2} \\
&= \frac{\pi}{\sin(\alpha\pi)} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz),
\end{aligned}$$

where the last step is the Beta integral which requires that $\alpha \in (0, 1)$. It remains to justify the two applications of the stochastic Fubini's theorem (see Theorem 5.30 of Chapter one in [Dal+09], or also [Wal86] or Theorem 4.18 of [DZ92]) in (3.10) and (3.11) in the following two steps.

Step 1. In this step, we justify the change of orders in (3.10). It suffices to prove the following condition:

$$\begin{aligned}
I_1 &:= \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(s-r, y-z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, u(r, z_i)) \right) \\
&= \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y(s, y)\|_2^2 \\
&< +\infty.
\end{aligned}$$

But this follows immediately from (3.3). Indeed

$$\begin{aligned}
\int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y(s, y)\|_2^2 &\leq C \int_{\mathbb{R}^d} dy G(t-s, x-y) J_2^*(s, y) \mathcal{H}_\alpha(s) \\
&= C \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_{\mathbb{R}^d} \mu_2^*(dx) G(s, y-z) \\
&= C \mathcal{H}_\alpha(s) J_2^*(t, x).
\end{aligned}$$

Then Condition (1.19) implies that $I_1 < \infty$.

Step 2. Similarly, as for (3.11), we need to show that

$$\begin{aligned}
I_2 &:= \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(t-r, x-z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, u(r, z_i)) \right) < \infty.
\end{aligned}$$

By the Cauchy Schwartz inequality, (3.7) and because $\alpha \in (0, 1/2)$,

$$\begin{aligned} I_2 &\leq C \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x - z_i) J_2^*(r, z_i) \right) \\ &= C' \int_0^t dr (t-r)^{-\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x - z_i) J_2^*(r, z_i) \right). \end{aligned}$$

Now by the same arguments as those leading to (3.3) (with 2α there replaced by α), we see that

$$I_2 \leq C J_*^2(t, x) \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{r} |\xi|^2\right),$$

which is finite by (1.19) where we replace α with $\alpha/2$. Indeed, since $r(t-r) \geq tr/2$ for $r \in [0, t]$, then

$$\int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{r} |\xi|^2\right) \leq \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right) < \infty.$$

This completes the proof of Lemma 3.3. $\square$

We are now ready to prove part (b) of Theorem 1.5.

Proof of Theorem 1.5 (b). In this proof we will use $\stackrel{\mathcal{L}}{=}$ to denote equality in law, otherwise, the equality holds almost surely. Let $t_0 > 0$ and consider the following restarted equation:

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}_{t_0/2}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, \cdot) = u(t_0/2, \cdot), \end{cases} \quad (3.12)$$

E:SHErandin

where $\dot{W}_{t_0/2}(t, x) = \dot{W}(t + t_0/2, x)$ denotes the time shifted noise. We note that (1.16) implies that this random initial condition, $u(t_0/2, \cdot)$ satisfies (1.7) and thus the solution to (3.12) exists and we denote the solution as v . By the Markov property of the solution to (1.1), we may say that for any $t \geq 0$, $u(t + t_0/2, x) \stackrel{\mathcal{L}}{=} v(t, x)$, where $\mathcal{L}$ means the equality is in law. In other words, for all $t \geq 0$, the following holds:

$$u(t + t_0/2, x) \stackrel{\mathcal{L}}{=} v(t, x) = (G(t, \cdot) * u(t_0/2, \cdot))(x) + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) b(y, v(s, y)) W_{t_0/2}(ds, dy).$$

We will use the notation Y_v and $\mathcal{G}_{\hat{\rho}, q}^v$ where

$$Y_v(s, y) := \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0/2}(dr, dz),$$

and

$$\begin{aligned} \mathcal{G}_{\hat{\rho}, q}^v(s) &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|v(0, y)\|_q dy \right)^2 \hat{\rho}(x) dx \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|u(t_0/2, y)\|_q dy \right)^2 \hat{\rho}(x) dx. \end{aligned}$$

We define $\mathcal{G}_{\hat{\rho}, q}^u(s)$ and Y_u in an equivalent manner. We wish to be able to apply the factorization result (3.9) and also the bound in (3.5) for Y_v . This requires us to show that $v(0, \cdot) = u(t_0/2, \cdot)$

satisfies (1.14) and (3.4) with $\hat{\rho}$. Indeed, Hölder's inequality and the admissible weight property imply that

$$\begin{aligned}\mathcal{G}_{\hat{\rho},q}^v(s) &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|u(t_0/2, y)\|_q dy \right)^2 \hat{\rho}(x) dx \\ &\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G(s, x-y) \|u(t_0/2, y)\|_q^2 dy \hat{\rho}(x) dx \\ &\leq C(t_0) \int_{\mathbb{R}^d} \|u(t_0/2, y)\|_q^2 \hat{\rho}(y) dy.\end{aligned}$$

Now by (2.7),

$$\begin{aligned}C(t_0) \int_{\mathbb{R}^d} \|u(t_0/2, y)\|_q^2 \hat{\rho}(y) dy &\leq C(t_0) \int_{\mathbb{R}^d} \left(G(t_0/2, \cdot) * (1 + \|\mu\|_q) \right)^2(y) \hat{\rho}(y) dy \\ &= C(t_0) \mathcal{G}_{\hat{\rho},q}^u(t_0/2) < \infty,\end{aligned}$$

where the finiteness follows from (1.18). Finally, this and the continuity of $\mathcal{H}_\alpha(s)$ for $s \in [0, t_0/2]$ shows that,

$$\int_0^{t_0/2} (\mathcal{G}_{\hat{\rho},q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds \leq \mathcal{G}_{\hat{\rho},q}^u(t_0/2)^{q/2} \int_0^{t_0/2} (\mathcal{H}_\alpha(s))^{q/2} ds < \infty.$$

Thus we are justified in saying that by (3.9),

$$u(t_0, x) \stackrel{\mathcal{L}}{=} v(t_0/2, x) = \left(G(t_0/2, \cdot) * u(t_0/2, \cdot) \right)(x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y_v](t_0/2, x). \quad (3.13)$$

We now define the set $\mathcal{K}(\Lambda)$ as

$$\mathcal{K}(\Lambda) = \{ (G(t_0, \cdot) * y(\cdot))(x) + (F_\alpha h)(x), \quad \text{where } \|y\|_{\hat{\rho}} \leq \Lambda \text{ and } \|h\|_{L^q(0,1,L_{\hat{\rho}}^2)} \leq \Lambda \}.$$

Then by Proposition 3.1, $\mathcal{K}(\Lambda)$ is relatively compact in $L_{\hat{\rho}}^2$ for $\Lambda > 0$. Moreover, by (3.5) and Chebyshev's inequality we have

$$\begin{aligned}\mathbb{P}\{u(t_0, \cdot) \notin \mathcal{K}(\Lambda)\} &= \mathbb{P}\{v(t_0/2, \cdot) \notin \mathcal{K}(\Lambda)\} \\ &\leq \mathbb{P} \left[\left(\int_0^{t_0/2} \|Y_v(s, \cdot)\|_{\hat{\rho}}^q ds \right)^{1/q} > \frac{\pi\Lambda}{\sin(\alpha\pi)} \right] + \mathbb{P} \left[\|u(t_0/2, \cdot)\|_{\hat{\rho}} > \Lambda \right] \\ &\leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \mathbb{E} \int_0^{t_0/2} \|Y_v(s, \cdot)\|_{\hat{\rho}}^q ds + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right) \\ &\leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \Theta \int_0^{t_0/2} (\mathcal{G}_{\hat{\rho},q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right) < \infty\end{aligned} \quad (3.14)$$

With all this said, (3.14) implies that

$$\mathbb{P}\{u(t_0, \cdot) \in \mathcal{K}(\Lambda)\} \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \left(\Theta \int_0^{t_0/2} (\mathcal{G}_{\hat{\rho},q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds \right) - \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right)$$

Therefore, we can choose $\Lambda > 0$ big enough such that

$$\frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \left(\Theta \int_0^{t_0/2} (\mathcal{G}_{\hat{\rho},q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds \right) + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right) < \epsilon,$$

then we get that

$$\mathbb{P}\{u(t_0, \cdot) \in \mathcal{K}(\Lambda)\} \geq 1 - \epsilon.$$

We can now deduce from this and the semigroup property of the solution to the heat equation that

$$\mathbb{P}\{u(t, \cdot) \in \mathcal{K}(\Lambda)\} \geq (1 - \epsilon)\mathbb{P}\{\|u(t - t_0/2, \cdot)\|_{\hat{\rho}} < \Lambda\}.$$

We now apply Chebyshev's inequality and (1.18) to see that for all $t > t_0/2$ that there exists a $C > 0$ such that

$$\mathbb{P}\{\|u(t - t_0/2, \cdot)\|_{\hat{\rho}} \geq \Lambda\} \leq \frac{\mathbb{E}\left(\|u(t - t_0/2, \cdot)\|_{\hat{\rho}}^2\right)}{\Lambda^2} \leq C \frac{\mathcal{G}_{\hat{\rho}, q}\left(t - \frac{t_0}{2}\right)}{\Lambda^2} \leq \frac{C}{\Lambda^2}.$$

Thus, by enlarging Λ if necessary,

$$\mathbb{P}\{\|u(t - t_0/2, \cdot)\|_{\hat{\rho}} \geq \Lambda\} \leq \epsilon, \quad \forall t \geq t_0/2,$$

thus completing the proof of (1.21). $\square$

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